

විශ්වවිද්‍යාල පාලන අධිකාරිය - කොළඹ අධ්‍යාපන කලාපය மேல் மாகாணக் கல்வித் திணைக்களம் - கொழும்பு கல்வி வலையம் Western Provincial Education Department - Colombo Educational Zone		
දෙවන වාර ඇගයීම - 2016 இரண்டாம் தவணைமதிப்பீடு - 2016 Second Term Evaluation - 2016		
10 ශ්‍රේණිය தரம் 10 Grade 10	විද්‍යාව I விஞ்ஞானம்/வினாத்தாள் - I Science Paper I	එක වැඩි இரண்டுமணித்தியாலங்கள் One Hour

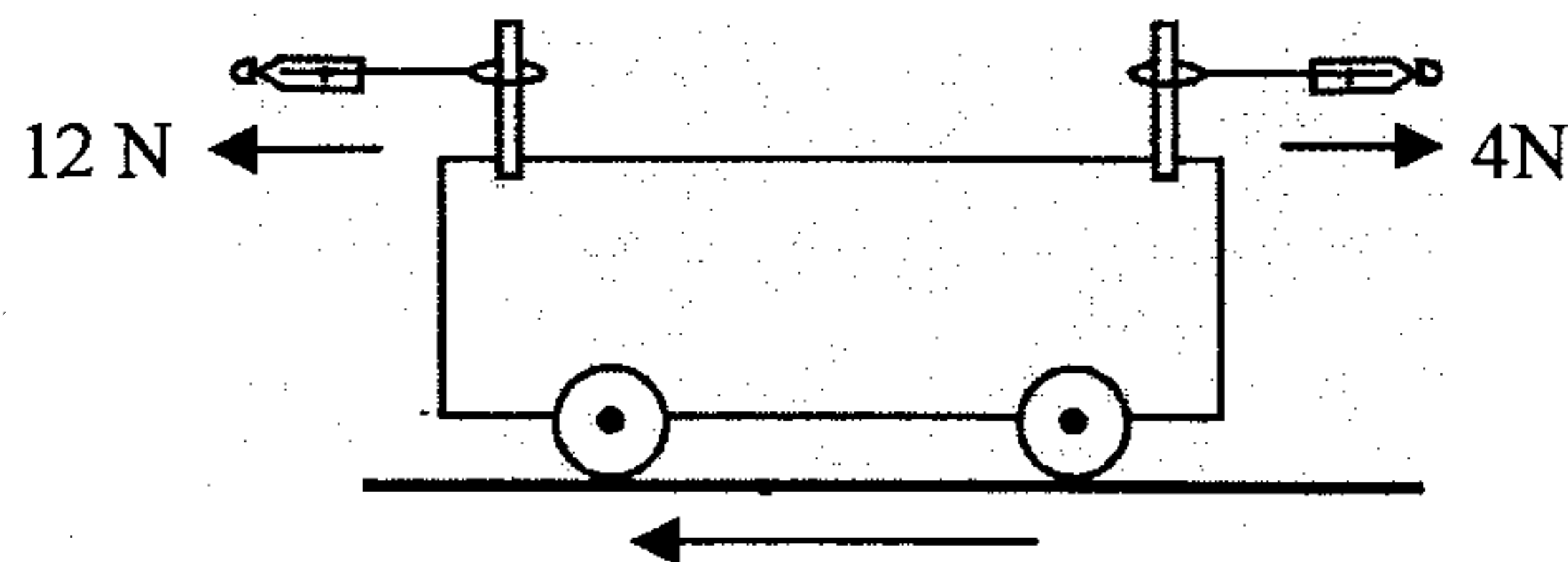
- Note: (i) Answer all questions.
(ii) In each of the questions 1 to 40, pick one of the alternatives (1), (2), (3), (4) which you consider is correct or most appropriate .
(iii) Mark a cross (x) on the number corresponding to your choice in the answer sheet provided.

- Element that is needed for transmission of nerve impulses is,
(1) K (2) Fe (3) I (4) N
- Electronic configuration of $^{27}_{13}\text{Al}$ is,
(1) 2, 8, 7 (2) 2, 8, 3 (3) 2, 8, 17 (4) 2, 8, 8, 9
- S. I. unit of measuring displacement is,
(1) ms^{-1} (2) ms^{-2} (3) m (4) m^{-1}s
- An example for disaccharides is,
(1) lactose (2) glucose (3) starch (4) galactose
- Scientist who introduced the planetary model of the atom is,
(1) Neils Bohr (2) Ernest Rutherford (3) J.J. Thompson (4) Dimithri Mendaleev
- Mass of an object is 6 kg. Momentum of the object when it moves at a velocity of 50 ms^{-1} is,
(1) 300 kg (2) $(50/6) \text{ ms}^{-1}/\text{kg}$ (3) $(6/50) \text{ kg/ms}^{-1}$ (4) 300 kgms^{-1}
- Deficiency of which vitamin causes slow-down in blood clotting ?
(1) C (2) D (3) K (4) E
- Atomic number of a certain element is 17. Mass number of it is 37. That element is,
(1) a mono – valent metal (2) a mono – valent non-metal
(3) an inert gas (4) a bivalent element.
- An object was projectd upward with an initial velocity of 60 ms^{-1} . If the gravitational acceleration is 10 ms^{-2} time taken by it to reach the maximum height in seconds is,
(1) 5 (2) 10 (3) 6 (4) 60
- Network of flat or tubular sacs present within the cytoplasm of cells is,
(1) ribosome (2) golgi complex (3) mitochondrium (4) endoplasmic reticulum

(11) How is it prevented the attraction of negatively charged electrons towards the positively charged nucleus?

- (1) as positive and negative charges repel each other
- (2) as electrons are attracted towards energy levels
- (3) as electrons rotate at a high speed around the nucleus
- (4) as there are neutrons in the nucleus which are neutral

(12) Following diagram illustrates the application of two forces on a trolley kept on a smooth surface. Resultant force applied on the trolley is,



- (1) 16 N
- (2) 8 N
- (3) 16 N
- (4) 8 N

(13) An aqueous solution of a saccharide was heated with few drops of dilute sulphuric acid and Benedic solution. Then a brick red precipitate was appeared. The saccharide present in solution is,

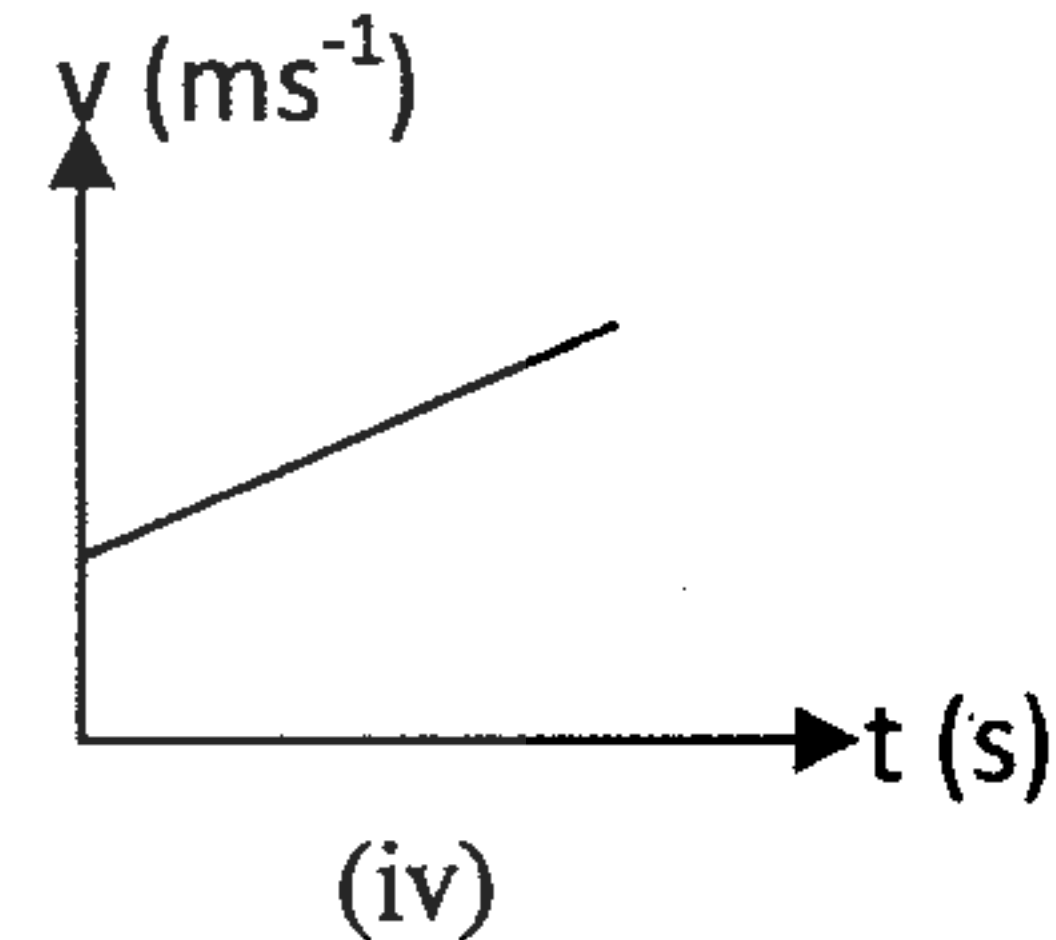
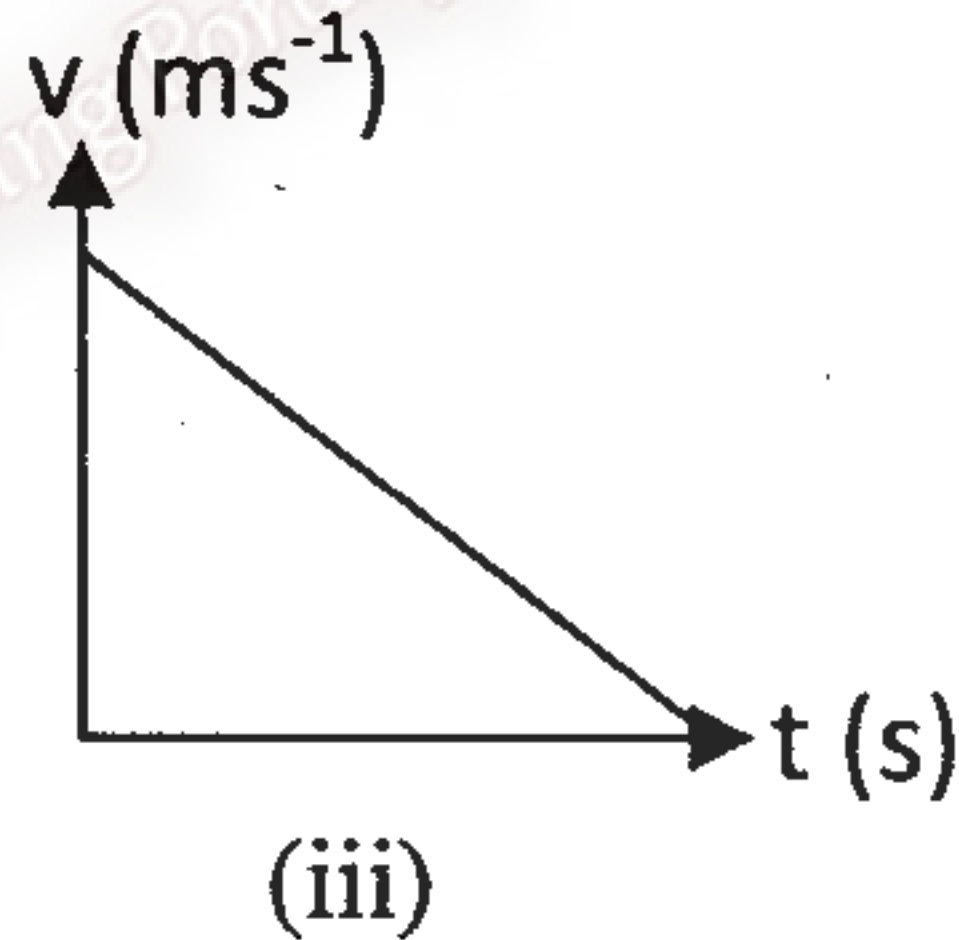
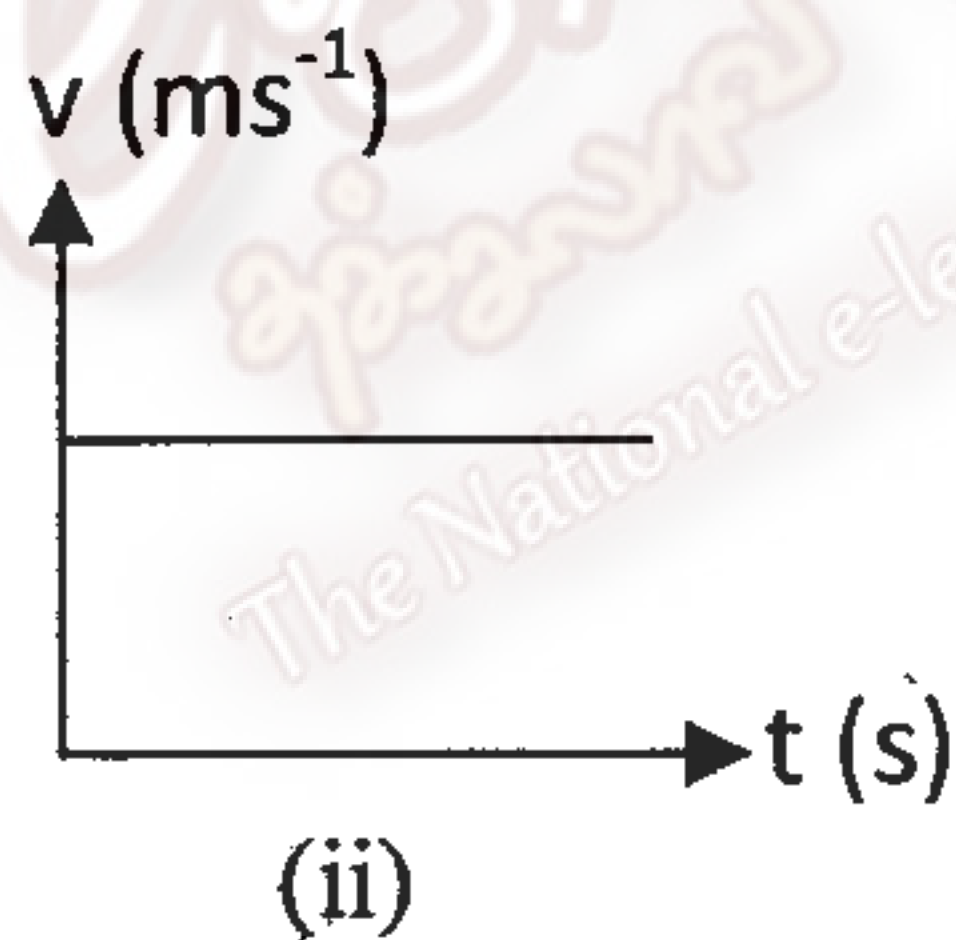
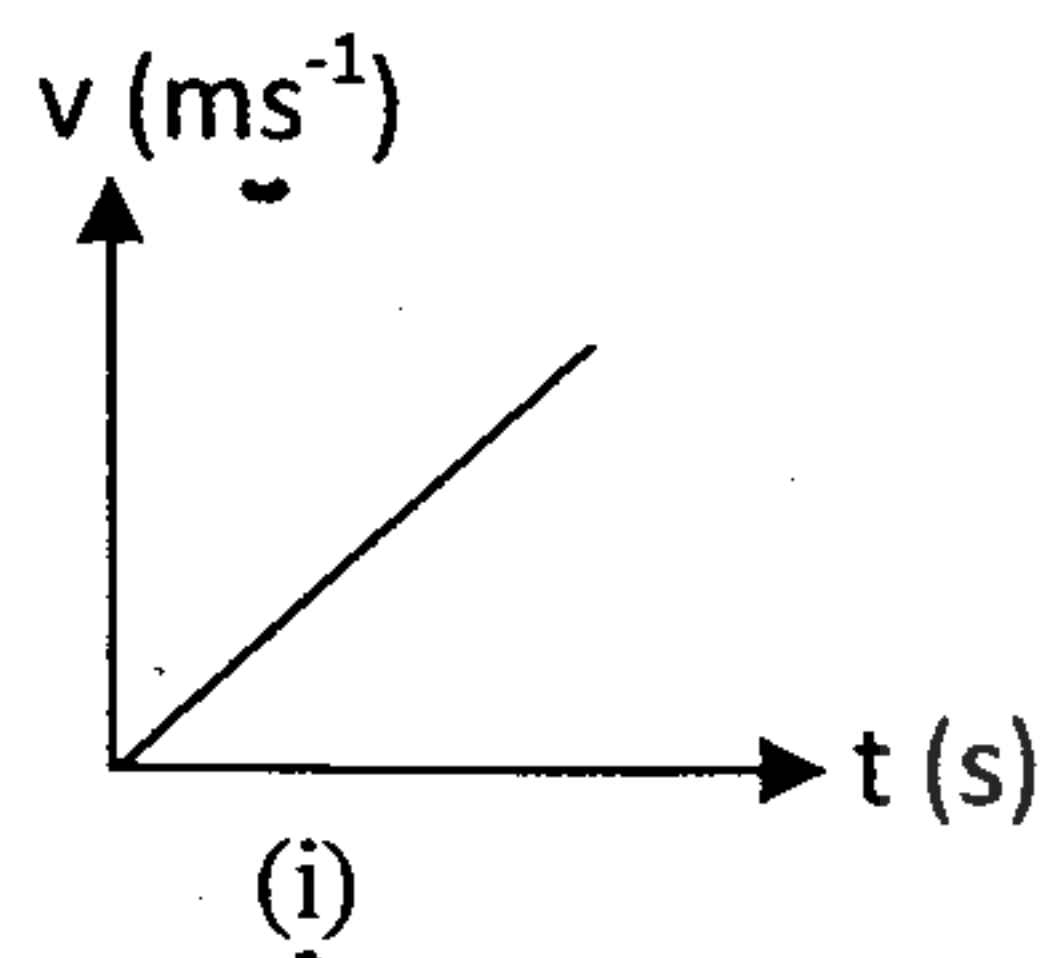
- (1) glucose
- (2) fructose
- (3) sucrose
- (4) galactose

(14) Molecular formula of glucose is $C_6H_{12}O_6$. What is the mase of a one mole of glucose molecules?

(C= 12, O=16, H=1)

- (1) 180 g
- (2) 90 g
- (3) 270 g
- (4) 45 g

(15) Velocity – time graph showing the free-fall of a fruit from a tree is;



(16) The common carbohydrate present in animal liver and muscles is,

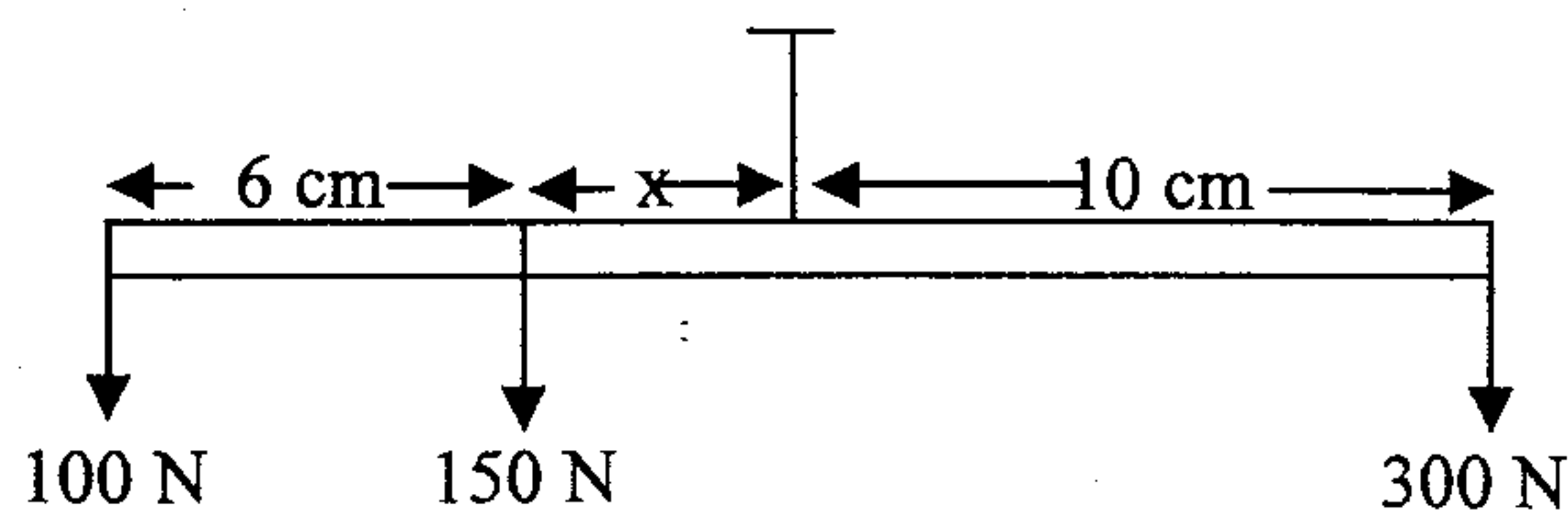
- (1) starch
- (2) glucose
- (3) glycogen
- (4) fructose

(17) Correct statement / statements regarding the valency shell of an atom is/are,

- a – they are shells with electrons
- b – it is the last shell with electrons
- c – it is the last shell with electrons that participate in bonding

- (1) a and b only
- (2) b and c only
- (3) only b
- (4) only c

(18)



If the above light rod is at equilibrium, length of 'X' is,

- (1) 10 cm
- (2) 12 cm
- (3) 9.6 cm
- (4) 3.2 cm

(19) This is not an example for under-ground stems,

- (1) ginger (2) onion (3) potato (4) manioc

(20) Chemical formula of sulphate of element A is ASO_4 . Chemical formula of its nitrate is,

- (1) $ANNO_3$ (2) A_2NO_3 (3) $A_2(NO_3)_2$ (4) $A(NO_3)_2$

(21) Which one of the following is not an application of Newton's third law?

- (1) rowing a boat (2) motion of a helicopter
(3) motion of a hydrogen balloon (4) motion of an air cracker

(22) An example for fungus is,

- (1) yeast (2) paramecium (3) chlamydomonas (4) ulva

(23) This is not an application of the nitrogen gas,

- (1) to prepare ammonia (2) to fill filamentous bulbs
(3) to fill food packets (4) to make pharmaceutical drugs

(24) Mass of an object is 2 kg. An unbalanced force (F) applied on it increases the velocity from 10 ms^{-1} to 20 ms^{-1} in 5 seconds. The value of F is,

- (1) 2 N (2) 4 N (3) 20 N (4) 0 N

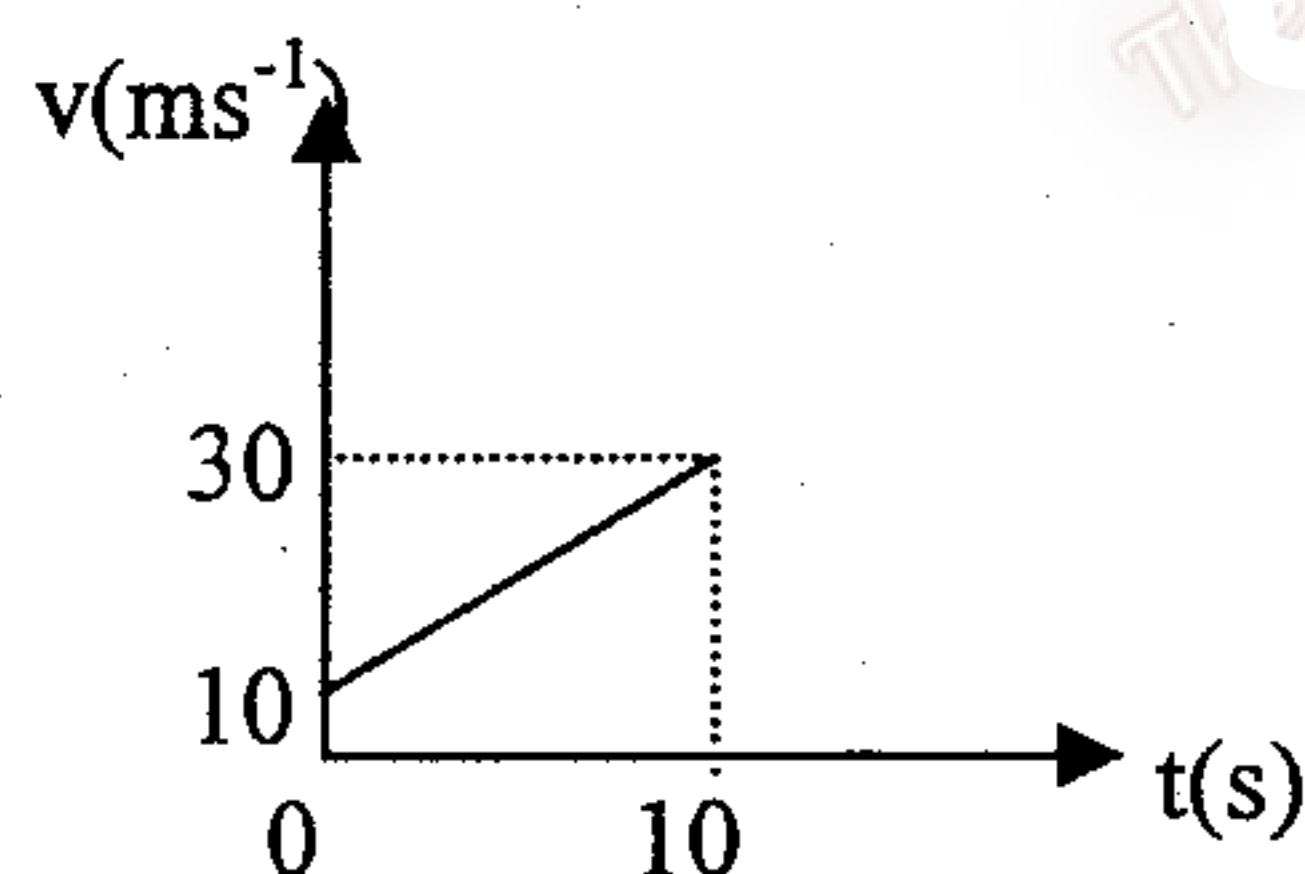
(25) Which of the following answers show the combination pattern of nitrogenous bases in nucleotides which are building units of nucleic acid?

- (1) A and C (2) T and C (3) T and G (4) A and T

(26) Substance that conducts electricity is,

- (1) crystals of common salt (2) aqueous copper sulphate solution
(3) distilled water (4) aqueous sugar solution

(27)



Velocity-time graph of an object is given. Displacement of that object is,

- (1) 100 m (2) 150 m
(3) 200 m (4) 300 m

(28) A certain somatic cell contains 10 pair of chromosomes. Number of chromosome pairs in daughter cells when that cell undergoes meiosis and mitosis respectively are,

- (1) 10, 10 (2) 10, 5 (3) 5, 10 (4) 10, 20

(29) When an ion is formed from a neutral atom,

- (1) number of electrons in the atom change (2) number of protons in the atom change
(3) number of neutrons in the atom change (4) number of protons and electrons in the atom change

(30) This is not an example for the moment of a couple of forces,

- (1) steering wheel (2) water pump (3) key (4) scissor

(31) Group of animals which are homeothermic are,

- (1) reptiles and birds (2) birds and mammals (3) amphibians and fish (4) reptiles and mammals

(32) Set of elements with the low electro-negativity is,

- (1) O, N, K (2) Mg, Ca, F (3) Cl, K, B (4) K, Ca, Na

(33) Static frictional force is defined as,

- (1) maximum frictional force applied on an object
(2) frictional force applied on a moving object
(3) frictional force applied on an object at rest, yet external force is applied.
(4) frictional force applied on an object when no forces are applied on it

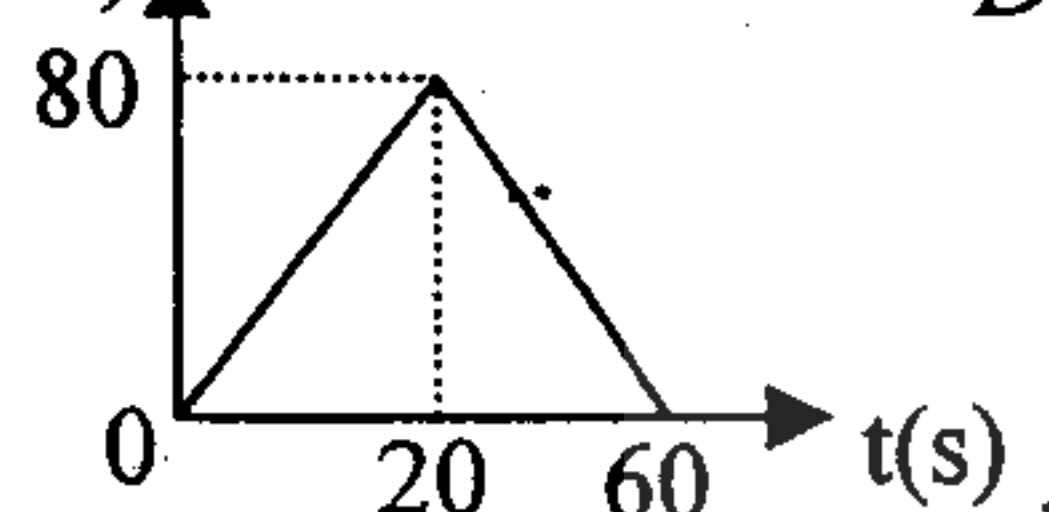
(34) Which of the following is not a function of plasma membrane?

- (1) acts as a cover for the cell (2) allows water and some other ions to enter the cell
(3) performs many metabolic activities (4) controls the removal of substances from the cell

(35) Number of molecules in 2 moles of carbon dioxide is,

- (1) 6.022×10^{23} (2) $2 \times 6.022 \times 10^{23}$ (3) 6.022×10^{24} (4) 3.011×10^{23}

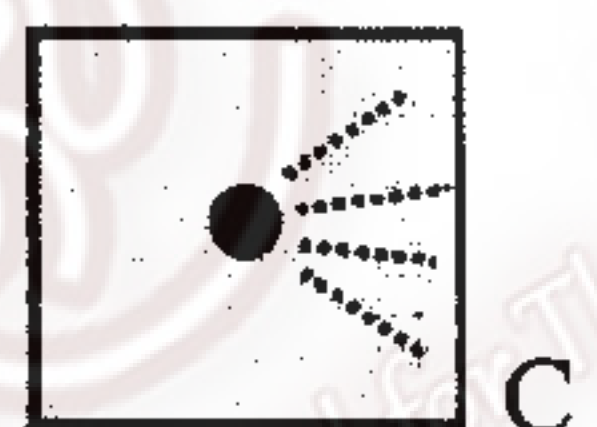
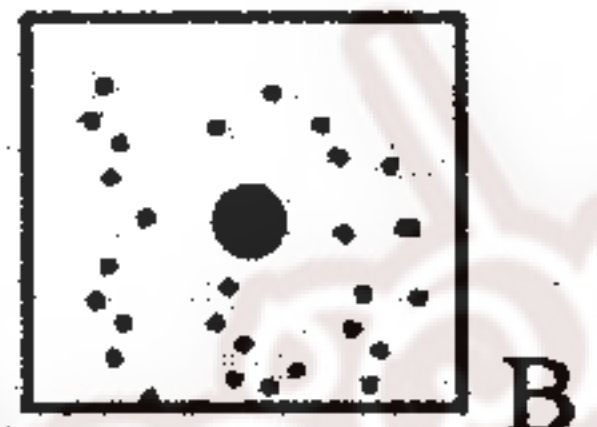
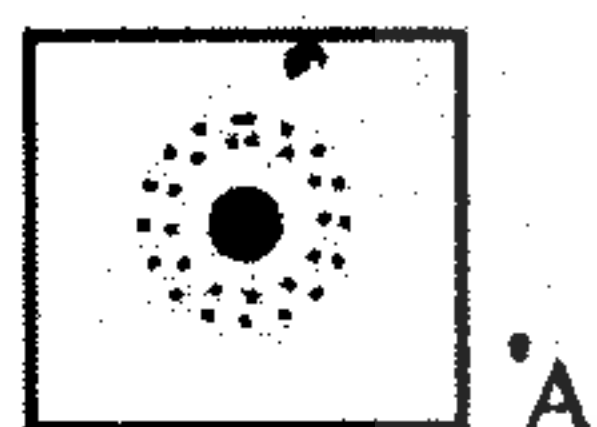
(36) $v(\text{ms}^{-1})$ Deceleration of the object given in the graph is,



- (1) -2 ms^{-2} (2) 4 ms^{-2}
(3) 2 ms^{-2} (4) $4/3 \text{ ms}^{-2}$

(37) Following diagram illustrate the fruit and seed dispersals of A, B and C plants.

(small dots represent the fruits and seeds)



According to the above diagrams their mechanisms of fruit and seed dispersal respectively are,

- (1) by animals, explosive mechanism, wind (2) by explosive mechanism, animals, wind
(3) by wind, animals, explosive mechanism (4) by animals, wind, explosive mechanism

(38) Correct statement regarding the covalent compounds is,

- (1) always they exist as solids (2) always they have low melting and boiling point
(3) certain covalent compounds dissolve in water (4) they consist of oppositely charged ions

(39) Some statements regarding cross-pollination are given below,

- a – it helps to form new variations among the plants
b – it does not support for the evolution process
c – it is a step in the sexual reproduction of plants

Correct statement/statements out of the above are,

- (1) a and b only (2) b and c only (3) a and c only (4) a, b, c all

(40) An equilibrium is represented by a set of forces on an object is,

